

DOMBAAS, Norway

WMO: 012330

Lat: 62.072N

Lon: 9.116E

Elev: 638

StdP: 93.89

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.7	-19.2	-24.1	0.5	-21.1	-21.7	0.6	-18.7	12.5	-4.8	11.0	-4.3	0.6	250	0.703

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	23.4	13.9	21.2	13.5	19.1	12.5	15.5	20.2	14.5	18.9	13.5	17.8	3.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.6	10.5	17.4	12.6	9.8	16.3	11.6	9.2	15.0	45.6	20.4	42.7	19.0	39.8	18.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	7.9	6.7	-23.2	25.8	2.9	2.2	-25.3	27.4	-27.0	28.7	-28.7	30.0	-30.8	31.6		
(5)			-23.5	16.8	2.9	1.3	-25.5	17.7	-27.2	18.4	-28.8	19.1	-30.9	20.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.7	-7.3	-5.8	-2.2	2.2	7.1	10.7	13.7	11.9	8.2	2.3	-2.9	-5.8		
	DBStd	8.40	5.74	5.65	4.41	3.76	4.21	3.16	3.24	2.56	3.05	3.64	4.78	6.12		
	HDD10.0	2934	537	444	378	234	112	28	5	10	69	238	388	491		
	HDD18.3	5705	796	677	636	484	348	230	148	201	303	496	638	749		
	CDD10.0	276	0	0	0	1	22	48	119	68	17	1	0	0		
	CDD18.3	4	0	0	0	0	0	0	4	0	0	0	0	0		
	CDH23.3	57	0	0	0	0	2	9	46	0	0	0	0	0		
	CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0		
	WSAvg	2.6	2.6	2.7	2.9	2.7	2.7	2.6	2.5	2.4	2.6	2.6	2.3	2.5		
(15) Precipitation	PrecAvg	431	33	26	22	17	31	47	66	63	30	33	32	32		
	PrecMax	599	75	54	57	39	104	119	123	118	68	86	89	70		
	PrecMin	336	2	2	3	3	7	5	28	24	6	3	5	10		
	PrecStd	57	19	12	13	9	21	25	25	24	14	17	20	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.6	7.3	9.6	14.7	23.0	24.7	26.8	22.1	18.5	14.3	9.5	6.7		
		MCWB	2.8	3.9	4.5	7.9	12.9	13.2	15.6	14.1	12.6	11.0	6.2	3.9		
	2%	DB	3.5	4.6	7.2	12.5	19.6	22.1	24.6	20.0	16.5	10.8	6.5	4.9		
		MCWB	1.6	1.8	3.9	6.5	11.1	12.5	15.1	13.6	11.4	7.4	4.7	2.5		
	5%	DB	1.7	2.9	5.3	10.4	17.3	19.0	22.3	18.2	14.7	9.0	4.5	3.3		
		MCWB	0.1	0.7	2.3	5.4	10.2	11.6	14.4	13.3	10.6	6.5	2.9	1.3		
	10%	DB	0.1	1.1	3.7	8.5	14.7	16.8	20.0	16.6	13.2	7.4	2.8	1.6		
		MCWB	-1.1	-0.7	1.0	4.4	9.2	10.5	13.5	12.3	9.8	5.1	1.5	0.0		
	0.4%	WB	3.2	4.0	5.6	8.7	13.7	14.6	17.7	16.0	13.7	11.8	6.4	4.1		
		MCDB	5.3	7.0	8.3	14.1	20.7	19.7	24.6	19.3	17.1	13.9	8.9	6.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	1.7	2.0	3.8	7.1	12.3	13.6	16.2	14.9	12.2	8.1	4.8	2.6		
		MCDB	3.3	3.9	6.8	11.5	17.6	19.4	21.8	18.1	15.3	10.2	6.5	4.5		
	5%	WB	0.3	0.8	2.6	5.8	10.8	12.6	15.2	13.9	11.2	6.6	3.1	1.4		
		MCDB	1.5	2.8	5.1	9.8	15.9	17.8	20.2	17.1	14.1	8.7	4.4	3.0		
	10%	WB	-1.1	-0.5	1.2	4.5	9.5	11.3	14.1	12.8	10.0	5.3	1.5	0.1		
		MCDB	0.1	1.1	3.4	8.2	14.3	15.8	18.9	15.9	12.8	7.3	2.6	1.4		
	5% DB	MDBR	5.4	6.0	7.4	8.2	8.9	9.3	8.9	8.0	7.6	5.8	4.9	5.4		
		MCDBR	6.4	6.7	8.5	11.2	12.9	13.9	12.8	10.2	9.9	7.4	6.2	6.9		
		MCWBR	5.0	5.1	5.8	6.5	6.5	6.6	5.7	5.4	5.9	5.3	4.9	5.4		
		MCWBR	6.2	6.5	8.0	9.8	11.2	10.9	10.9	8.9	8.8	6.7	6.1	6.8		
(40) Clear-Sky Solar Irradiance	taub		0.199	0.250	0.284	0.315	0.325	0.325	0.337	0.327	0.295	0.275	0.208	0.169		
		taud	2.076	2.236	2.217	2.209	2.339	2.459	2.449	2.470	2.540	2.430	2.110	1.955		
	Ebn at Noon		522	734	829	871	894	901	878	858	830	705	492	353		
		Edh at Noon	33	61	90	113	108	98	97	87	67	51	31	20		
(44) All-Sky Solar Radiation	RadAvg		0.25	0.90	2.38	4.31	5.21	5.08	4.67	3.63	2.30	1.04	0.33	0.14		
	RadStd		0.04	0.15	0.27	0.34	0.41	0.41	0.46	0.29	0.22	0.12	0.05	0.01		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-120	-144	N/A	N/A

Nomenclature: See separate page